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Solid-state imaging device

Abstract

A solid-state imaging device includes an N-type semiconductor layer, an element layer including a photoelectric conversion element and an active element, an interconnect layer providing an interconnect for the active element, and an element isolation trench penetrating the semiconductor layer. The element layer includes a P-type region and an N-type region. A first hole storage layer is formed on a surface of the semiconductor layer on a side opposite to the element layer. A second hole storage layer is formed in contact portions of the semiconductor layer and the element layer with the element isolation trench. The P-type region of the element layer and the first hole storage layer are connected to each other by the second hole storage layer.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a solid-state imaging device.
BACKGROUND ART
(2) In recent years, a lot of image sensors with greatly improved sensitivities in an infrared region have been proposed. Since silicon has a lower infrared absorption sensitivity, extending the distance through which infrared rays pass through a silicon substrate is effective in improving the sensitivity. For this reason, a lot of techniques of forming a thick silicon substrate have been proposed.
(3) On the other hand, a backside illumination image sensor has been proposed as a technique of improving the sensitivity. This image sensor allows the entry of light from the surface opposite to a surface on which an active element is formed. Patent Document 1 discloses a technique of forming a p-type layer on the back surface and fixing the potential via a deep p-type well from the front surface to reduce a dark current on the back surface.
CITATION LIST
Patent Document
(4) Patent Document 1: Japanese Unexamined Patent Publication No. 2003-31785
SUMMARY OF THE INVENTION
Technical Problem
(5) However, with an increase in the thickness of the silicon substrate to improve the infrared sensitivity, extending the deep p-type well region to the back surface becomes difficult. As a result, the potential of the p-type layer on the back surface cannot be fixed, whereby increasing the dark current.
(6) In view of the forgoing, it is an objective of the present disclosure to provide an imaging device capable of providing excellent image quality without degrading a dark current even with an increase in the thickness of a silicon substrate of a backside illumination image sensor to improve infrared sensitivity.
Solution to the Problem
(7) A solid-state imaging device according to the present disclosure includes a pixel array of a plurality of unit pixels. Each of the plurality of unit pixels includes a photoelectric conversion element that generates a signal charge by photoelectric conversion, and an active element that converts the signal charge into an electric signal and outputs the electric signal. The solid-state imaging device includes: an N-type semiconductor layer; an element layer stacked on the semiconductor layer and including the photoelectric conversion element and the active element; an interconnect layer stacked on the element layer and providing an interconnect for the active element; and an element isolation trench penetrating the semiconductor layer. The element layer includes a P-type region and an N-type region. A first hole storage layer is formed on a surface of the semiconductor layer on a side opposite to the element layer. A second hole storage layer is formed in contact portions of the semiconductor layer and the element layer with the element isolation trench. The P-type region of the element layer and the first hole storage layer are connected to each other by the second hole storage layer.
Advantages of the Invention
(8) The solid-state imaging device according to the present disclosure reduces an increase in a dark current even with an increase in the thickness.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic cross-sectional view of a solid-state imaging device according to a first

embodiment.
(2) FIG. **2** is a diagram illustrating a solid-state imaging device according to a comparative example.
(3) FIG. **3** is a schematic cross-sectional view of a solid-state imaging device according to a second embodiment.
(4) FIG. **4** includes plan views schematically illustrating shapes of a near-via DTI region in the solid-state imaging device in FIG. **3**.
(5) FIG. **5** is a schematic cross-sectional view of a solid-state imaging device according to a third embodiment.
(6) FIG. **6** includes plan views schematically illustrating shapes of a near-edge DTI region in the solid-state imaging device in FIG. **5**.
(7) FIG. **7** is a schematic cross-sectional view of a solid-state imaging device according to a fourth embodiment.
(8) FIG. **8** includes plan views schematically illustrating shapes of an end n-type well region and a near-edge DTI region in the solid-state imaging device in FIG. **7**.
(9) FIG. **9** is a plan view schematically illustrating a configuration of a solid-state imaging device according to a fifth embodiment.
(10) FIG. **10** is a diagram for explaining a method of manufacturing a solid-state imaging device according to the present disclosure.
(11) FIG. **11** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **10**.
(12) FIG. **12** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **11**.
(13) FIG. **13** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **12**.
(14) FIG. **14** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **13**.
(15) FIG. **15** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **14**.
(16) FIG. **16** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **15**.
(17) FIG. **17** is a diagram for explaining the method of manufacturing the solid-state imaging device according to the present disclosure, following FIG. **16**.

DESCRIPTION OF EMBODIMENTS

(18) Embodiments of the present disclosure will be described below with reference to the drawings.

First Embodiment

(19) FIG. **1** schematically shows a cross section of a main part of an exemplary solid-state imaging device **100** according to a first embodiment of the present disclosure.
(20) The solid-state imaging device **100** includes an N-type semiconductor layer **101**, an element layer **102** stacked thereon and including a photoelectric conversion element and an active element, and an interconnect layer **103** stacked thereon and providing an interconnect for the active element and other elements. In the solid-state imaging device **100**, incident light enters the element layer **102** from the side close to the N-type semiconductor layer **101**, that is, opposite to the interconnect layer **103**.
(21) The N-type semiconductor layer **101** and the element layer **102** are formed using an N-type epitaxial layer on an N-type semiconductor substrate (not shown). The element layer **102** is obtained by forming P-type and N-type regions and an insulating layer in an upper region of the N-type epitaxial layer. A region of the N-type epitaxial layer below the element layer **102** serves as the N-type semiconductor layer **101**. This is an example, and the method of forming the N-type semiconductor layer **101** and the element layer **102** is not particularly limited.

(22) The element layer **102**, more specifically, a P-type well region **110**, an N-type well region **111**, a P-type layer **112**, photodiodes **113** which are N-type regions, a high-concentration N-type layer **115**, and a high-concentration P-type layer **116** are formed by impurity injection, for example. In addition, shallow trench isolation (STI) regions **114** which are element isolation trenches are obtained by forming trenches penetrating the N-type semiconductor layer **101** and removing parts of the element layer **102**, and filling the trenches with impurities.
(23) The photodiodes **113** perform photoelectric conversion using incident light and generate signal charges. The active element including the N-type well region **111** and other regions converts the signal charges generated by the photodiodes **113** into electric signals, and outputs the electric signals. Unit pixels include the photodiodes **113** and the active element. The unit pixels are arranged in a matrix into a pixel array. In addition, a circuit isolated by the P-type well region and the N-type well region is provided on a periphery of the pixel array.
(24) Located on the element layer **102** is the interconnect layer **103** including an on-element insulating film **120** and a transfer gate **121** on the on-element insulating film **120**.
(25) A deep trench isolation (DTI) region **135** which is an element isolation trench penetrating the N-type semiconductor layer **101** is provided. The DTI region **135** reaches the P-type layer **112** of the element layer **102** from the surface opposite to the element layer **102**. The DTI region **135** is obtained by forming a trench **136** with a depth that allows penetration through the N-type semiconductor layer **101** and cutting of part of the element layer **102**, and filling the inside of the trench **136** with the first insulating film **132**.
(26) Before filling with the first insulating film **132**, a metal oxide film **131** made of hafnium-dioxide (HfO.sub.2) or aluminum-oxide (Al.sub.2O.sub.3) is formed to cover the sidewalls and the bottom surface of the trench **136**.
(27) The metal oxide film **131** covers the inside of the trench **136** and the surface of the N-type semiconductor layer **101**. Accordingly, the first insulating film **132** fills the inside of the trench **136** with the metal oxide film **131** interposed therebetween. The first insulating film **132** covers the metal oxide film **131** outside the trench **136**, as well. A second insulating film **133** covers the insulating film **132**.
(28) In the metal oxide film **131**, negative fixed charge is generated on a side facing the N-type semiconductor layer **101** and the element layer **102**. Accordingly, holes are supplied from a GND terminal through the P-type well region **110** to contact portions of the N-type semiconductor layer **101** and the element layer **102** with the metal oxide film **131**, where the holes are stored to form a P-type hole storage layer **130**. More specifically, a first hole storage layer **130***a* is formed along the surface of the N-type semiconductor layer **101** on the side opposite to the element layer **102**, and a second hole storage layer **130***b* is formed on the bottom surface and sidewall of the trench **136**.
(29) A via (i.e., a through-silicon via (TSV)) **140** penetrating the N-type semiconductor layer **101** and the element layer **102** is formed. The via **140** is insulated from the N-type semiconductor layer **101** and the element layer **102** by the second insulating film **133**. The via **140** is connected to a copper interconnect **122** on the side closer to the interconnect layer **103**, and to an electrode pad **141** on the side closer to the N-type semiconductor layer **101**.
(30) In the solid-state imaging device **100** according to the present disclosure described above, the P-type region (e.g., the P-type layer **112**) in the element layer **102** and the first hole storage layer **130***a* which is the P-type region on the side of the N-type semiconductor layer **101** opposite to the element layer **102** are electrically connected to each other by the second hole storage layer **130***b* along the DTI region **135**. Even with an increase in the thickness of the N-type semiconductor layer **101**, holes are thus supplied through the second hole storage layer **130***b* on the sidewalls of the DTI region **135**, thereby making it possible to reduce an increase in a dark current.
(31) FIG. **2** shows a solid-state imaging device **100***a* according to a comparative example, which corresponds to the solid-state imaging device **100** without the DTI region **135**. If the solid-state imaging device **100***a* includes an N-type semiconductor layer **101** with a sufficiently small

thickness T, a deep P-type well region can be formed from the side closer to the element layer **102** by impurity injection or other methods, and extend to the surface (i.e., the back surface) opposite to the element layer **102**. In this case, the potential of the P-type layer on the back surface can be fixed.

(32) However, with an increase in the thickness T of the N-type semiconductor layer **101** to improve infrared sensitivity, forming the P-type well region to the back surface becomes difficult. As a result, the potential of the P-type layer on the back surface cannot be fixed, thereby increasing a dark current.

(33) By contrast, in the solid-state imaging device **100** according to the present disclosure shown in FIG. **1**, the DTI region **135** is formed from the back surface, and the hole storage layer **130** is formed in the contact portion with the DTI region **135**. Even with an increase in the thickness of the N-type semiconductor layer **101**, holes are thus supplied through the hole storage layer **130** to the back surface. As a result, an increase in the dark current is reduced, and a captured image has an improved quality.

Second Embodiment

(34) FIG. **3** is a schematic cross-sectional view of a solid-state imaging device **100***b* according to a second embodiment. The solid-state imaging device **100***b* is the same as the solid-state imaging device **100** in FIG. **1** except that the solid-state imaging device **100***b* includes a near-via DTI region **135***a* around the via **140**. Similarly to the DTI region **135** of the solid-state imaging device **100**, the near-via DTI region **135***a* includes an insulating film filling the trench with the metal oxide film **131** interposed therebetween.

(35) FIG. **4** shows, as a plan view, a schematic planar configuration A of the via **140** and the near-via DTI region **135***a* of the solid-state imaging device **100***b*. As shown here, the near-via DTI region **135***a* surrounds the periphery of the via **140**.

(36) The via **140** serves as a TSV penetrating the silicon substrate (or the N-type semiconductor layer **101**) to form the electrode pad **141** on the side of the N-type semiconductor layer **101** opposite to the element layer **102**. The DTI region **135***a* includes, on a side surface, a p-n junction, which causes a leakage current.

(37) To address this, the near-via DTI region **135***a* surrounds the via **140** in the solid-state imaging device **100***b* according to this embodiment. Accordingly, an N-type semiconductor layer **101***a* around the via **140** is electrically isolated, which reduces the generation of the leakage currents.

(38) A planar configuration B in FIG. **4** is a variation of the layout of the near-via DTI region **135***a*. In the planar configuration A, the near-via DTI region **135***a* has a rectangular shape with each corner bending at an angle of 90°. On the other hand, the planar configuration B has a shape (i.e., an octagon) with each corner bending twice at an obtuse angle (135° in this example) in place of bending each corner of the rectangle at an angle of 90°. The configuration without any part bending at 90° in this manner provides a stable size of the near-via DTI region **135***a* and stable filling characteristics of the first insulating film **132** or other films, when forming the near-via DTI region **135***a*. The shape of the near-via DTI region **135***a* in the plan view is not limited to the octagon shown and may be another shape.

Third Embodiment

(39) FIG. **5** is a schematic cross-sectional view of an exemplary solid-state imaging device **100***c* according to a third embodiment. As compared to the solid-state imaging device **100** shown in FIG. **1**, the solid-state imaging device **100***c* includes none of the via **140**, and the copper interconnect **122** and electrode pad **141** connected to the via **140**. The solid-state imaging device **100***c* further includes a near-edge DTI region **135***b* with the same or similar structure to that of the DTI region **135**.

(40) FIG. **6** shows, as a plan view, a schematic planar configuration C of the near-edge DTI region **135***b* of the solid-state imaging device **100***c*, and an internal circuit region **151** surrounded by the near-edge DTI region **135***b*. The internal circuit region **151** includes both an N-type well region and

a P-type well region to serve as a photoelectric conversion element and an active element, for example.

(41) In FIG. **5**, the left end of the solid-state imaging device **100***c* is a chip end **150** of the solid-state imaging device **100***c* divided (diced) as a chip. The chip end **150** includes p-n junctions at two points, namely, between the P-type well region **110** and the N-type semiconductor layer **101** and between the N-type semiconductor layer **101** and the hole inducing layer **130***a*. These p-n junctions, too, cause a leakage current. To address this, the near-edge DTI region **135***b* is formed in the solid-state imaging device **100***c* at the peripheral edge of the chip so as to be along the chip end **150**. The near-edge DTI region **135***b* continuously surrounds the inside region. Accordingly, an N-type semiconductor layer **101***b* near the chip end **150** is electrically isolated, thereby making it possible to reduce an increase in a leakage current.

(42) In a variation, similarly to the near-via DTI region **135***a* shown in the planar configuration B of FIG. **4**, the near-edge DTI region **135***b*, too, may include no part bending at an angle of 90° (may include a part bending at an obtuse angle of, for example, 135°). This is shown as a planar configuration D of FIG. **6**. Accordingly, the near-edge DTI region **135***b* can be formed stably.

Fourth Embodiment

(43) FIG. **7** is a schematic cross-sectional view of an exemplary solid-state imaging device **100***d* according to a fourth embodiment. Unlike the solid-state imaging device **100***c* of FIG. **5**, the solid-state imaging device **100***d* includes, in the element layer **102**, an end N-type well region **117** near the chip end **150**. The end N-type well region **117** reaches the N-type semiconductor layer **101**. In this configuration, the p-n junctions at the chip end **150** (or the diced surface) are reduced to only one p-n junction between the N-type semiconductor layer **101** and the hole inducing layer **130***a*, which further reduces leakage currents.

(44) FIG. **8** shows, as a plan view, the end N-type well region **117** near the chip end **150**, the P-type well region **110** inside the end N-type well region **117**, the near-edge DTI region **135***b* inside the P-type well region **110** and along the chip end **150**, and the internal circuit region **151** more inward than the near-edge DTI region **135***b*. In a variation, the near-edge DTI region **135***b* may bend at an obtuse angle (135° here) as in the planar configuration B of FIG. **4**. This is shown as a planar configuration F of FIG. **8**.

(45) The configuration with fewer p-n junctions using the N-type well region in the element layer **102** so as to reach the N-type semiconductor layer **101** is applicable to a region other than the vicinity of the chip end **150**. For example, an N-type well region as described above may be formed around the via **140** shown in FIG. **3** (not shown). This configuration can reduce the number of the p-n junctions around the via **140** and reduce the leakage currents.

Fifth Embodiment

(46) FIG. **9** is a schematic plan view of an exemplary solid-state imaging device **100***e* according to a fifth embodiment. FIG. **9** shows the entire chip of the solid-state imaging device **100***e*.

(47) In FIG. **9**, a P-type well region **110**, an N-type well region **111**, and a via **140** are provided. As described in the second embodiment (FIGS. **3** and **4**), the via **140** and the near-via DTI region **135***a* surrounding the via **140** are also provided. This configuration reduces leakage currents around the via **140**. In addition, as described in the third embodiment (FIGS. **5** and **6**), the near-edge DTI region **135***b* is provided along the chip end **150**. This configuration reduces leakage currents near the chip end **150**.

(48) The solid-state imaging device **100***e* includes, around the center of the chip, a pixel array **152** where a plurality of unit pixels are arranged in a matrix. An annular near-array DTI region **135***c* surrounds the pixel array **152**. The near-array DTI region **135***c* has the same or similar structure to that of the DTI region **135** shown in FIG. **1**, for example.

(49) The inside of the pixel array **152** can be electrically isolated from the periphery, using the P-type well region **110** and the near-array DTI region **135***c*. It is therefore possible to reduce effects of noise of a peripheral circuit on the pixel array **152**, for example.

Sixth Embodiment

(50) Next, a method of manufacturing a solid-state imaging device will be described as a sixth embodiment. In particular, a method of manufacturing the DTI region **135** will be described in detail. FIGS. **10** to **17** are diagrams for explaining the method of manufacturing the solid-state imaging device according to the present disclosure. In these figures, the solid-state imaging device is shown upside down with respect to FIG. **1** and other figures.

(51) FIG. **10** shows a stage before forming the DTI region **135**. In order to obtain this structure, first, the N-type epitaxial layer is formed on the N-type semiconductor substrate. By impurity injection or other means, the P-type well region **110**, the N-type well region **111**, the P-type layer **112**, the N-type photodiodes **113**, and other elements are formed in the upper region of the epitaxial layer. The STI regions **114** are also formed. Accordingly, the pixel array **152** including the photoelectric conversion element, the active element, and other elements is formed so that the epitaxial layer serves as the element layer **102**. The region of the N-type epitaxial layer under the element layer **102** serves as the N-type semiconductor layer **101**.

(52) Subsequently, the interconnect layer **103** obtained by embedding a plurality of interconnects **161** in an insulating layer **162** is formed on the element layer **102**. Further, another wafer is, as a support substrate **160**, bonded onto the interconnect layer **103**. Next, the initially used N-type semiconductor substrate is thinned and removed. As necessary, the N-type semiconductor layer **101** may also be thinned to have a predetermined thickness. FIG. **10** shows this state with the support substrate **160** located below.

(53) Next, the step shown in FIG. **11** is performed. Here, a trench **136** for forming the DTI region **135** is formed. The trench **136** is formed to penetrate the N-type semiconductor layer **101** and to cut part of the element layer **102**. For example, a mask having an opening at a predetermined position may be provided by lithography, and the trench **136** may be formed by etching to a necessary depth using the mask.

(54) Next, the step shown in FIG. **12** is performed. Here, the metal oxide film **131** is formed to cover the bottom and sidewalls of the trench **136** and cover the N-type semiconductor layer **101** (the surface opposite to the element layer **102**). At this time, the trench **136** is not filled completely, and a space is left. The metal oxide film **131** is specifically a HfO.sub.2 film or an Al.sub.2O.sub.3 film formed by a method such as chemical vapor deposition (CVD).

(55) Once the metal oxide film **131** is formed, holes are induced and stored in contact portions of the N-type semiconductor layer **101** and the element layer **102** with the metal oxide film **131**, thereby forming the P-type hole storage layer **130**. Accordingly, the metal oxide film **131** covers the inside of the trench **136** and the N-type semiconductor layer **101**, with the hole storage layer **130** interposed therebetween.

(56) Next, the step in FIG. **13** will be described. Here, the first insulating film **132** is formed to fill the space left in the trench **136** and cover the metal oxide film **131** outside the trench **136**. For example, the first insulating film **132** may be made of an oxide film and formed by a method such as CVD. The DTI region **135** is formed in this manner.

(57) Next, the step in FIG. **14** will be described. Here, a through-hole **140***a* for forming the via **140** is formed. The through-hole **140***a* is formed to penetrate the N-type semiconductor layer **101** and the element layer **102** and partially cut the first insulating film **132** and the insulating layer **162** of the interconnect layer **103** to reach one of the interconnects **161**. For this purpose, etching or other processing may be performed, for example. The interconnect **161** reached by the through-hole **140***a* corresponds to the copper interconnect **122** in FIG. **1**.

(58) The second insulating film **133** is formed to cover the sidewalls and the bottom surface of the through-hole **140***a* and cover the first insulating film **132** outside the through-hole **140***a*. After that, the second insulating film **133** is partially removed at the bottom of the through-hole **140***a* to expose the one of the interconnects **161**.

(59) Next, the step in FIG. **15** will be described. Here, in order to form the via **140**, a copper layer

140*b* is formed to fill the inside of the through-hole **140***a*. The copper layer **140***b* is formed to cover the second insulating film also outside the through-hole **140***a*. The copper layer **140***b* may be formed by plating, for example.
(60) Next, the step in FIG. **16** will be described. Here, the copper layer **140***b* outside the through-hole **140***a* is removed. For this, chemical mechanical polishing (CMP) may be used, for example. The via **140** obtained by filling the through-hole **140***a* with the copper layer **140***b* with the second insulating film **133** interposed therebetween is formed in this manner.
(61) Next, the step in FIG. **17** will be described. Here, the electrode pad **141** connected to the via **140** is formed on the side closer to the N-type semiconductor layer **101**. The electrode pad **141** may be made of aluminum, for example.
(62) The solid-state imaging device according to the present disclosure is manufactured in this manner. The materials, manufacturing method, shapes, and other features described above are mere examples, and the technique according to the present disclosure is not limited thereto.
INDUSTRIAL APPLICABILITY
(63) The solid-state imaging device according to the present disclosure reduces an increase in a dark current even with an increase in the thickness of the semiconductor layer, and is thus useful as a solid-state imaging device with an improved infrared sensitivity.
DESCRIPTION OF REFERENCE CHARACTERS
(64) **100** Solid-State Imaging Device **100***a* to **100***e* Solid-State Imaging Device **101** N-Type Semiconductor Layer **101***a* N-Type Semiconductor Layer **101***b* N-Type Semiconductor Layer **102** Element Layer **103** Interconnect Layer **110** P-Type Well Region **111** N-Type Well Region **112** P-Type Layer **113** Photodiode **114** STI **115** High-Concentration N-Type Layer **116** High-Concentration P-Type Layer **117** End N-Well Region **120** On-Element Insulating film **121** Electrode **122** Copper Interconnect **130** Hole Storage Layer **130***a* First Hole Storage Layer **130***b* Second Hole Storage Layer **131** Metal Oxide Film **132** First Insulating Film **133** Second Insulating Film **135** DTI Region **135***a* Near-Via DTI Region **135***b* Near-Edge DTI Region **135***c* Near-Array DTI Region **136** Trench **140** Via **140***a* Through-Hole **140***b* Copper Layer **141** Electrode Pad **150** Chip End **151** Internal Circuit Region **152** Pixel Array **160** Support Substrate **161** Interconnect **162** Insulating Layer

Claims

1. A solid-state imaging device including a pixel array of a plurality of unit pixels, each of the plurality of unit pixels including a photoelectric conversion element that generates a signal charge by photoelectric conversion, and an active element that converts the signal charge into an electric signal and outputs the electric signal, the solid-state imaging device comprising: an N-type semiconductor layer; an element layer stacked on the semiconductor layer and including the photoelectric conversion element and the active element; an interconnect layer stacked on the element layer and providing an interconnect for the active element; and an element isolation trench penetrating the semiconductor layer, wherein the element layer includes a P-type region and an N-type region, a first hole storage layer is formed on a surface of the semiconductor layer on a side opposite to the element layer, a second hole storage layer is formed in contact portions of the semiconductor layer and the element layer with the element isolation trench, the P-type region of the element layer and the first hole storage layer are connected to each other by the second hole storage layer, the solid-state imaging device is configured as a semiconductor chip, the element isolation trench includes, at a peripheral edge of the semiconductor chip, a near-edge element isolation trench that is continuous along an outer peripheral end, and the near-edge element isolation trench has a polygonal shape with an obtuse angle in a plan view.
2. A solid-state imaging device including a pixel array of a plurality of unit pixels, each of the plurality of unit pixels including a photoelectric conversion element that generates a signal charge

by photoelectric conversion, and an active element that converts the signal charge into an electric signal and outputs the electric signal, the solid-state imaging device comprising: an N-type semiconductor layer; an element layer stacked on the semiconductor layer and including the photoelectric conversion element and the active element; an interconnect layer stacked on the element layer and providing an interconnect for the active element; and an element isolation trench penetrating the semiconductor layer, wherein the element layer includes a P-type region and an N-type region, a first hole storage layer is formed on a surface of the semiconductor layer on a side opposite to the element layer, a second hole storage layer is formed in contact portions of the semiconductor layer and the element layer with the element isolation trench, the P-type region of the element layer and the first hole storage layer are connected to each other by the second hole storage layer, a P-type well region is formed on a periphery of the pixel array, the element isolation trench includes a near-array element isolation trench formed in the P-type well region and surrounding the pixel array, and the near-array element isolation trench surrounds the pixel array, in a polygonal shape with an obtuse angle in a plan view.